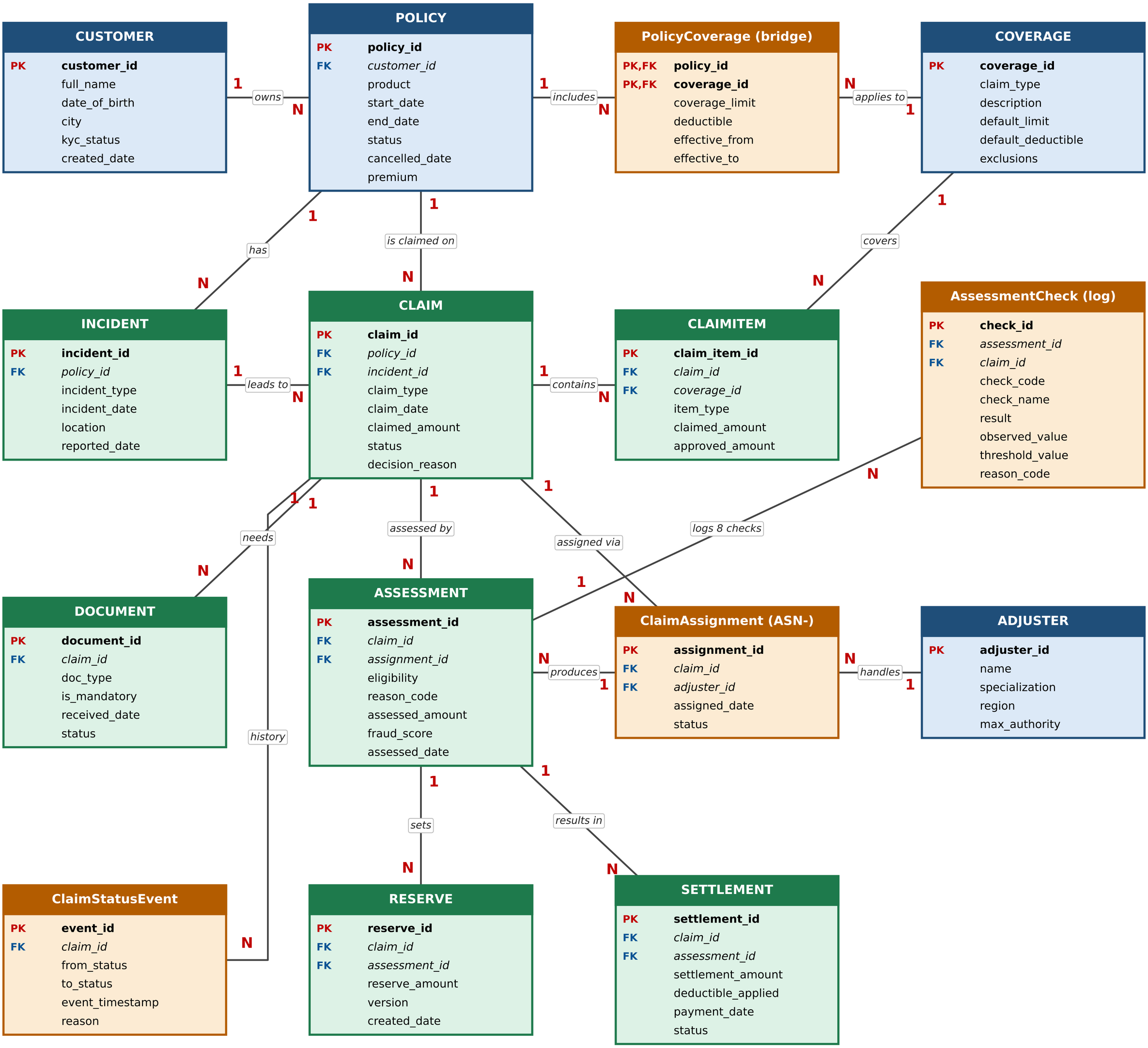
Insurance Claims Workflow - Entity Relationship Diagram

Synthetic data model for claim intake, coverage validation, assessment and settlement



LEGEND

Master table | Transaction table | Bridge / event-history table | PK = primary key | FK = foreign key | 1 / N = cardinality

- Cardinality: Customer 1:N Policy | Policy 1:N Claim | Claim 1:N ClaimItem | Policy N:M Coverage (bridge: PolicyCoverage) | Claim N:M Adjuster (bridge: ClaimAssignment)
- Optional links: Claim.incident_id may be empty for incomplete intake; Reserve is versioned (one claim can have several reserve records, history is never overwritten).
- AssessmentCheck stores the result of each eligibility check (CHK01 to CHK08), so every decision and its reason code can be traced.
- Settlement is 1:N per claim on purpose, so the 'claim paid twice' edge case can be represented and labelled.
- ID formats: POL-<product>-2026-000001 | CLM-2026-000001 | DOC-<claim-id>-001 | ASN-<claim-id>-001 | SET-<claim-id>-001
- ground_truth.csv is a separate file and is not part of this schema (no foreign keys point to it, so it can never leak into input features).